

Football Match Manager - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

Football Match Manager is a technical data model designed to manage the lifecycle of football competitions, focusing on the relational mapping between Leagues, Teams, Matches, and Users. The system implements a specific tracking pattern for match statuses and user-based match following, operating as a structured backend specification for sports data management.

1.2 Maturity Assessment

The project is currently in **REFINEMENT** status. While the data model is structurally complete in terms of entity mapping, the specifications lack a strategic framework (Goals, Scope) and contain unresolved technical uncertainties regarding password complexity and the precise scope of team name uniqueness.

1.3 Technical Stack

- **Primary Identifiers**: UUID (128-bit)
- **Temporal Standard**: UTC
- **Data Types**: DateTime, Enum, Boolean, Integer, String, URL

1.4 Architectural Constraints

- **Entity Integrity**: Match `homeTeamId` must not equal `awayTeamId`.
- **Temporal Logic**: Match `dateTime` must be in the future if status is 'upcoming'.
- **State-Based Requirements**: Match `homeScore` and `awayScore` must be non-null when status is 'finished'.
- **Value Constraints**: Scores must be ≥ 0 ; Team `foundedYear` must be between 1800 and the current year.
- **Uniqueness**: Team name must be globally unique; User email must be unique.
- **Relational Constraints**: `UserFollow` must maintain a composite unique constraint on (`userId`, `matchId`).
- **State Machine**: Match state transition flow: Upcoming $\rightarrow$ Live $\rightarrow$ Finished; Upcoming/Live $\rightarrow$ Postponed; Postponed $\rightarrow$ Live/Upcoming.
- **Character Limits**: Team name (100), Team abbreviation (5), Venue (200), User displayName (50), League name (100), League country (100).

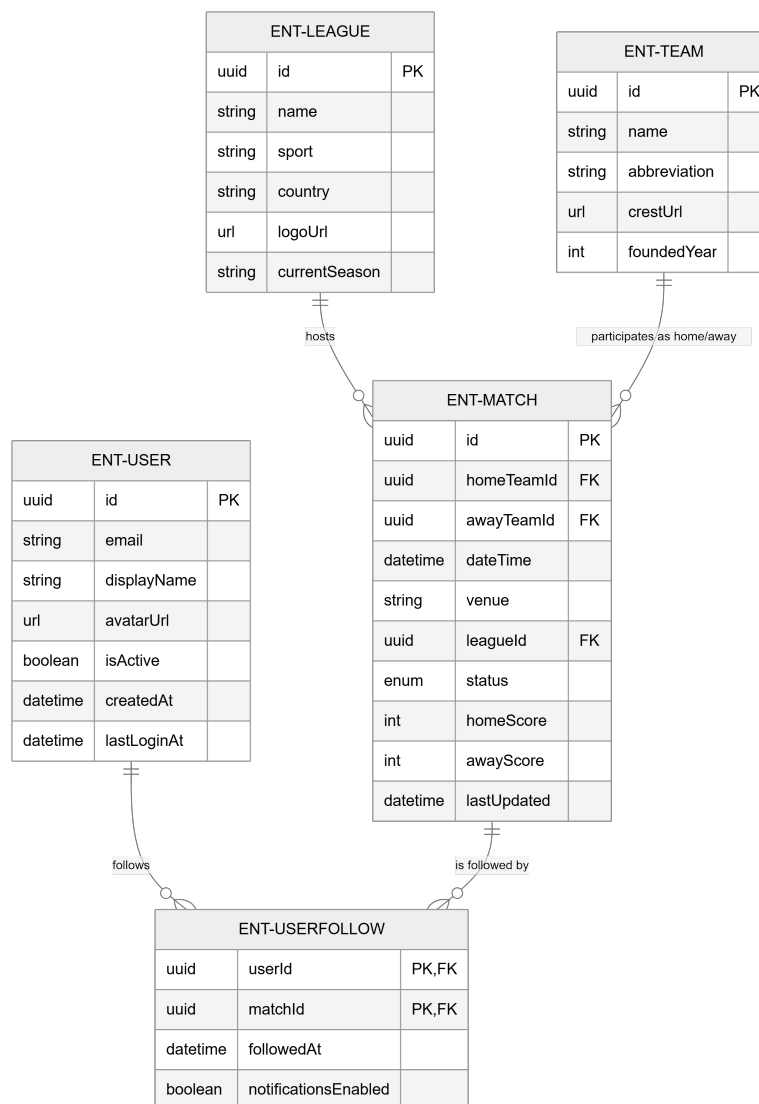
1.5 Critical Dependencies

- **Foreign Key Dependencies**:
 - `Match` $\rightarrow$ `League`, `Team`
 - `UserFollow` $\rightarrow$ `User`, `Match`
- **Performance Indexes**:

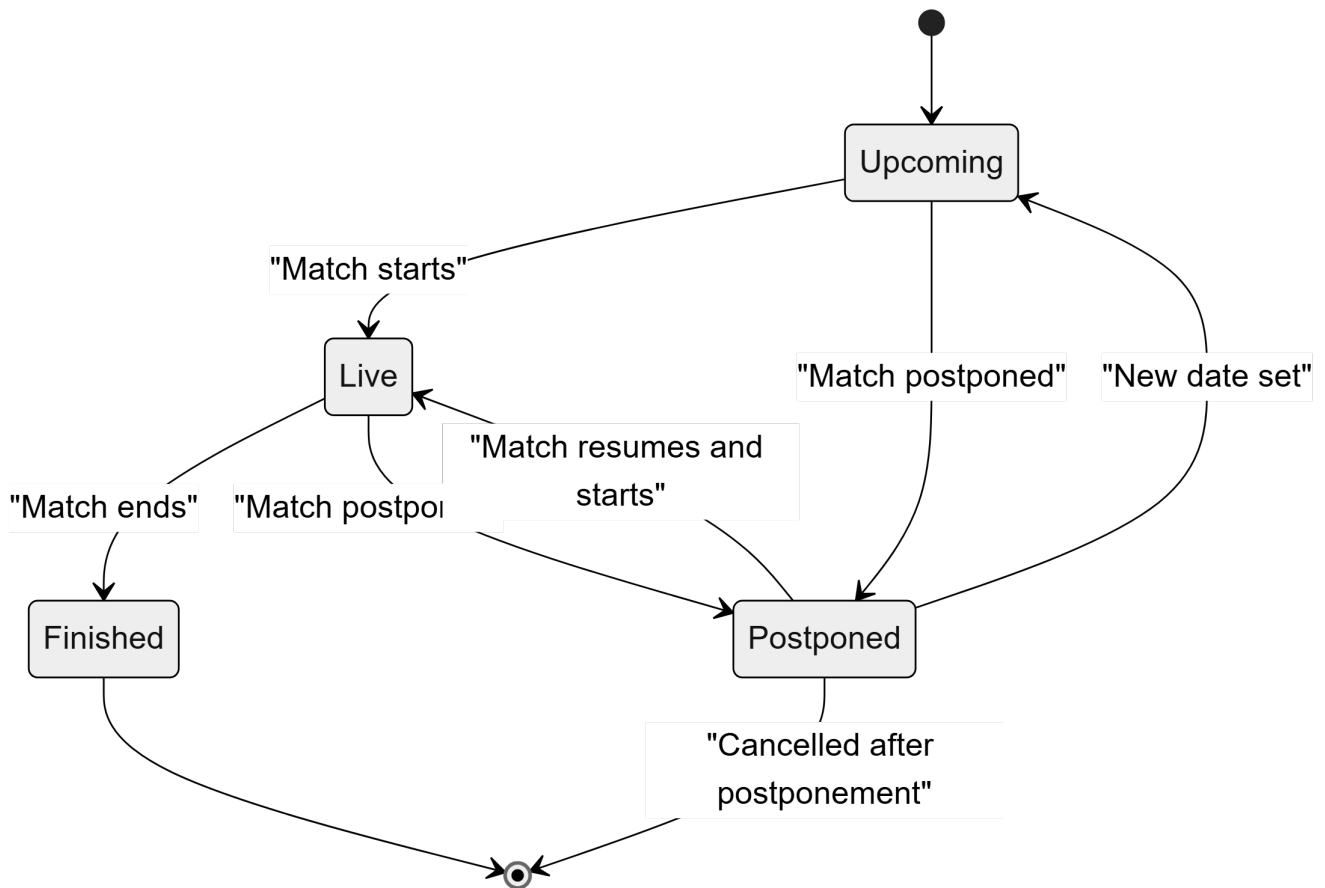
- `Match`: (`leagueId`, `dateTime`) and (`status`)
- `Team`: (`name`)
- `User`: (`email`)
- `UserFollow`: (`userId`) and (`matchId`)

2. Architecture Workflows & Visual Diagrams

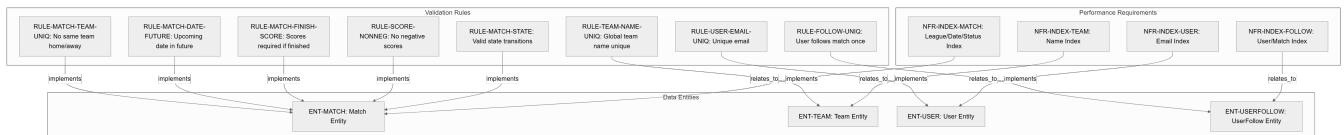
2.1 Entity Relationship Diagram



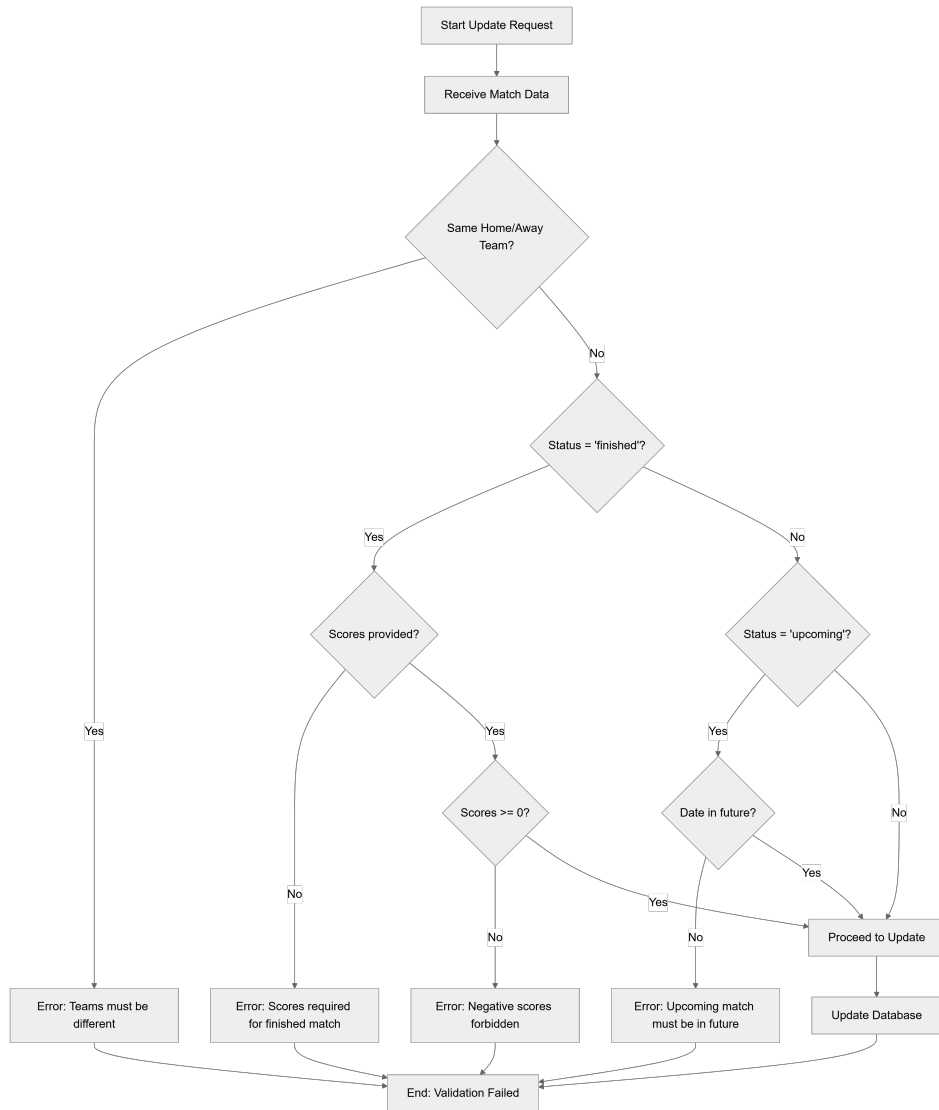
2.2 Match State Lifecycle



2.3 Requirements Traceability Matrix



2.4 Match Update Validation Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Requirement Type	Description	Target Entity
ENT-MATCH	Entity	Football match entity containing schedule, score and status.	N/A
ENT-TEAM	Entity	Football team entity containing identity and basic club info.	N/A
ENT-LEAGUE	Entity	Football league or tournament entity.	N/A
ENT-USER	Entity	Application user entity for authentication and personalization.	N/A
ENT-USERFOLLOW	Entity	Association entity between Users and Matches they follow.	N/A
RULE-MATCH-TEAM-UNIQUENESS	Functional	A match cannot have the same team as both home and away.	ENT-MATCH

RULE-MATCH-DATE-FUTURE	Functional	Match dateTime must be in the future for status 'upcoming'.	ENT-MATCH
RULE-MATCH-FINISH-S CORE	Functional	When status changes to 'finished', both homeScore and awayScore must be set (non-null).	ENT-MATCH
RULE-SCORE-NONNEG	Constraint	Scores cannot be negative.	ENT-MATCH
RULE-TEAM-NAME-UNI Q	Functional	Team name must be globally unique.	ENT-TEAM
RULE-USER-EMAIL-UNI Q	Functional	Email must be unique.	ENT-USER
RULE-FOLLOW-UNI Q	Constraint	A user can only follow a match once (Composite unique on userId, matchId).	ENT-USERFOLLOW
RULE-MATCH-STATE	Functional	Match state transitions: Upcoming $\rightarrow$ Live $\rightarrow$ Finished; Upcoming/Live $\rightarrow$ Postponed; Postponed $\rightarrow$ Live/Upcoming.	ENT-MATCH
NFR-INDEX-MATCH	Non-Functional	Indexes on (leagueId, dateTime) and (status) for Match performance.	ENT-MATCH
NFR-INDEX-TEAM	Non-Functional	Index on (name) for Team search.	ENT-TEAM
NFR-INDEX-USER	Non-Functional	Index on (email) for User authentication.	ENT-USER
NFR-INDEX-FOLLOW	Non-Functional	Indexes on (userId) and (matchId) for UserFollow retrieval.	ENT-USERFOLLOW

3.2 Security Rules

- **Authentication**: User email must be unique (`RULE-USER-EMAIL-UNI Q`).
- **Credential Policy**: Passwords must meet complexity requirements (minimum 8 characters), though specific detailed requirements are currently listed as an open question.
- **Account Status**: User activity is tracked via the `isActive` boolean flag.

3.3 Data Models

ENT-MATCH

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
homeTeamId	UUID	Reference to Team	Required, exists
awayTeamId	UUID	Reference to Team	Required, exists, $\neq$ homeTeamId
dateTime	DateTime	Match date and time (UTC)	Required, future if 'upcoming'
venue	String (200)	Stadium or location name	Optional
leagueId	UUID	Reference to League	Required, exists
status	Enum	upcoming, live, finished, postponed, cancelled	Required, default: upcoming

homeScore	Integer	Goals scored by home team	Optional, min 0, default null
awayScore	Integer	Goals scored by away team	Optional, min 0, default null
lastUpdated	DateTime	Timestamp of last update	Auto-set on update

ENT-TEAM

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	Team name	Required, globally unique
abbreviation	String (5)	Team abbreviation	Optional, unique per league
crestUrl	URL	URL to team crest/logo	Optional, valid URL
foundedYear	Integer	Year the club was founded	Optional, 1800 \$\le\$ year \$\le\$ current

ENT-LEAGUE

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	League name	Required
sport	String	Sport type	Required, default: "football"
country	String (100)	Country where league is based	Optional
logoUrl	URL	URL to league logo	Optional, valid URL
currentSeason	String	Current season identifier	Optional

ENT-USER

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
email	String (email)	User email address	Required, unique, valid format
displayName	String (50)	Display name for the user	Required
avatarUrl	URL	URL to user avatar/image	Optional, valid URL
isActive	Boolean	Whether the user account is active	Required, default: true
createdAt	DateTime	Account creation timestamp	Auto-set on create
lastLoginAt	DateTime	Last login timestamp	Updated on login

ENT-USERFOLLOW

Field	Type	Description	Validation
userId	UUID	Reference to User	Required, exists
matchId	UUID	Reference to Match	Required, exists

followedAt	DateTime	Timestamp of following	Required, auto-set on create
notificationsEnabled	Boolean	Notification preference	Optional, default: false

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	Add a section describing the purpose of the football match manager and its primary goals.
Scope & Out-of-Scope	MEDIUM	Define what the system will and will not do (e.g., no real-time betting, no complex player stats).
Open Questions & Uncertainties	LOW	Document any undecided parts of the data model, like the specific password complexity requirements.

4.2 Remediation & Workflow

1. ****Immediate Action****: Define the "Goals & Objectives" to align technical implementation with business value.
2. ****Refinement****: Resolve the open question regarding whether team name uniqueness is global or per-league.
3. ****Specification****: Formalize the password complexity requirements to complete the `ENT-USER` security profile.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Constraints	TECHNICAL_STACK	RULE-FOLLOW-UNIQ	Persistence layer rules enforcing data integrity, specifically including multi-column uniqueness requirements.
Cryptographic Hashing	TECHNICAL_STACK	RULE-USER-EMAIL-UNIQ	One-way transformation process required for securing stored credentials as implied by the password complexity mandate.
DateTime	TECHNICAL_STACK	ENT-MATCH	Temporal data primitive used for scheduling events and recording system-generated timestamps.

MUN	BUSINESS_DOMAIN	ENT-TEAM	An example of a short-form identifier for a sports club used to demonstrate the five-character limit.
Match Validation	BUSINESS_DOMAIN	Validation Rules	Logical checks ensuring no team plays itself, dates are chronologically correct for upcoming states, and scores are provided upon completion.
Relationships	BUSINESS_DOMAIN	ENT-MATCH	Structural associations between entities such as leagues, teams, and users.
Team Validation	BUSINESS_DOMAIN	RULE-TEAM-NAME-UNIQ	Enforcement of a globally distinct identifier for every registered sports organization.
UTC	TECHNICAL_STACK	ENT-MATCH	The standardized time reference used to synchronize match schedules across different geographic zones.
UUID	TECHNICAL_STACK	ENT-MATCH	128-bit unique identifier used as the primary key for all main entities to ensure global uniqueness.
User Validation	BUSINESS_DOMAIN	RULE-USER-EMAIL-UNIQ	Verification of distinct electronic mail addresses and strict adherence to credential complexity standards.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	An association entity that maps the interest of an account holder toward a specific athletic event.
UserFollow Validation	BUSINESS_DOMAIN	RULE-FOLLOW-UNIQ	The restriction preventing a single account from creating duplicate interest records for the same event.